



Digital Identity Verification

Group -15

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Problem Description

Current Situation:

- Pathao Pay verifies user identity using selfie-based liveness checks during account onboarding.

Emerging Threat:

- Advances in AI-generated images and deepfake technologies enable attackers to bypass basic liveness tests using synthetic or manipulated face images.



Problem Description

Core Problem:

- Existing verification methods struggle to reliably distinguish real human captures from AI-generated or machine-manipulated faces, especially under real-world conditions (compression, cropping, device variation).

Project Goal:

- Develop a robust AI-based fake face detection system capable of identifying synthetic or manipulated images and integrate it into Pathao Pay's verification pipeline.



Evolution of Identity Fraud



Evolution of Identity Fraud

①

Basic Spoofing

Simple methods like
printed photos
became common.

②

Human Review

Analysts began
manually reviewing
identity verification
cases.

③

AI-Generated Faces

Advanced technology
created hyper-
realistic synthetic
identities.



The Rise of Synthetic Faces

The emergence of **AI-generated faces** using GAN and diffusion models poses significant challenges for identity verification, undermining trust in digital systems and complicating security measures in financial transactions.

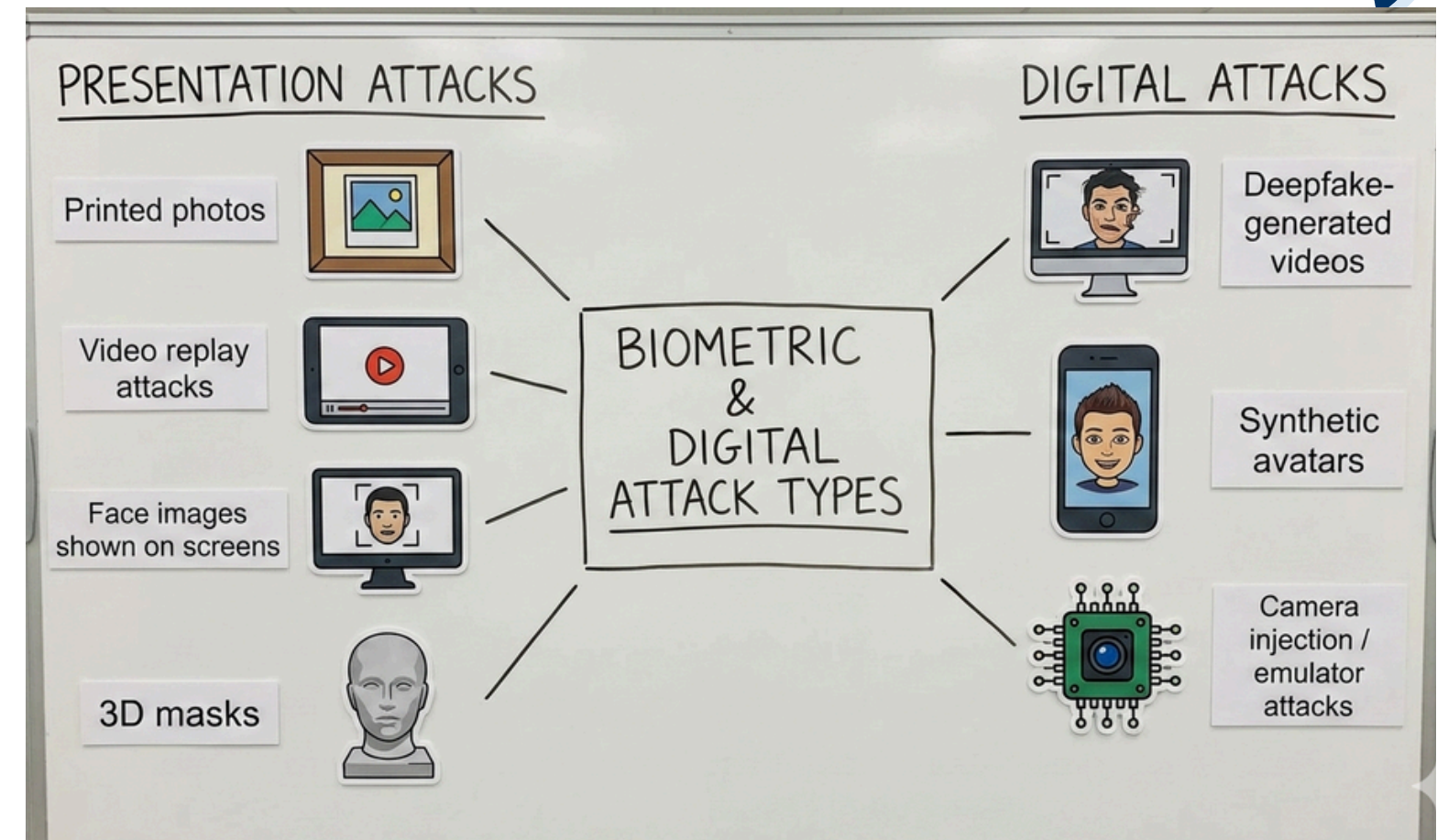
Common attack types

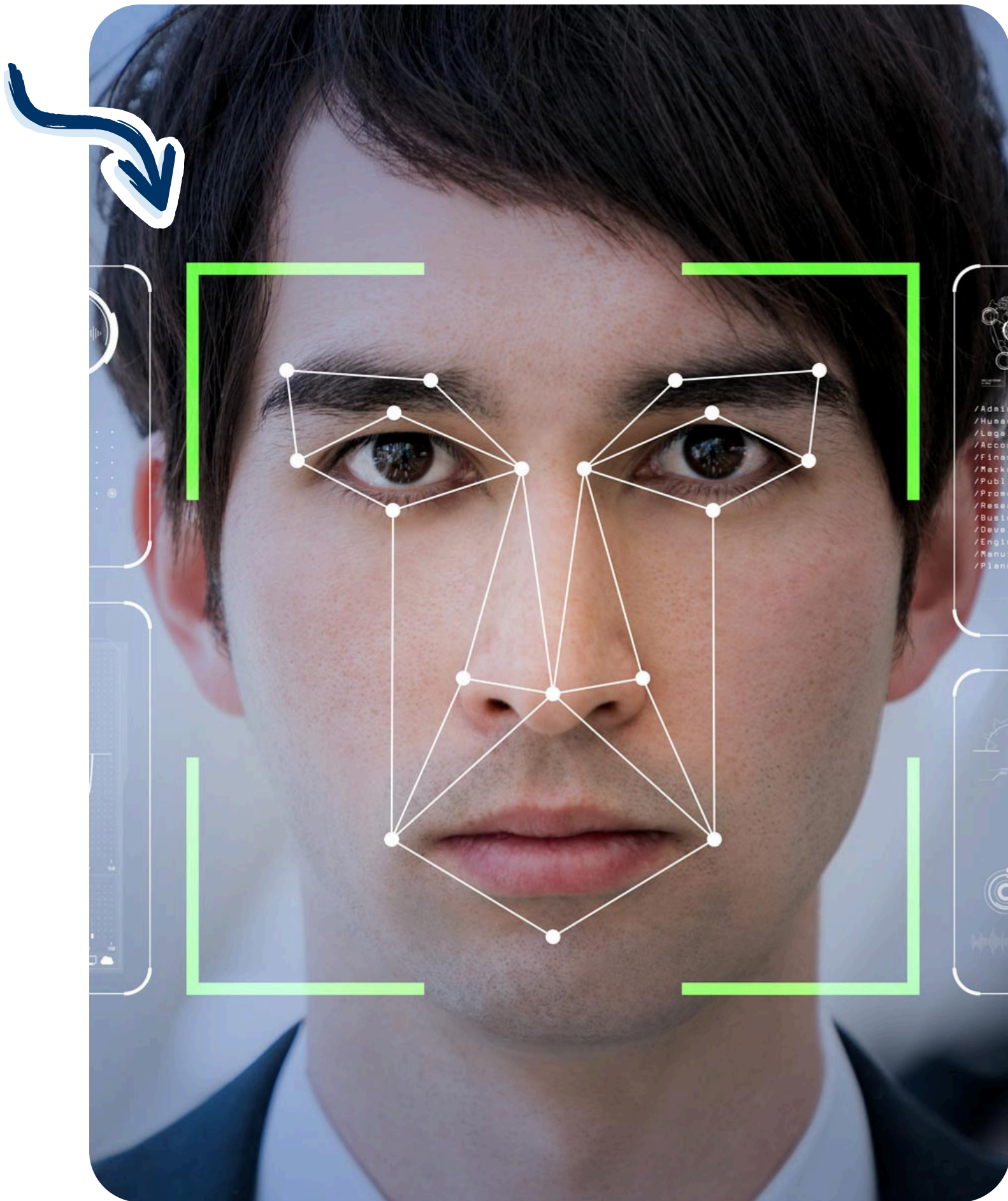
- **Presentation attacks**

- Printed photos
- Video replay attacks
- Face images shown on screens
- 3D masks

- **Digital attacks**

- Deepfake-generated videos
- Synthetic avatars
- Camera injection / emulator attacks





Detection Approaches

- **Active Liveness**

- User performs prompted actions (turn head, blink, smile)
- Effective against static attacks
- Vulnerable to video replay and deepfakes

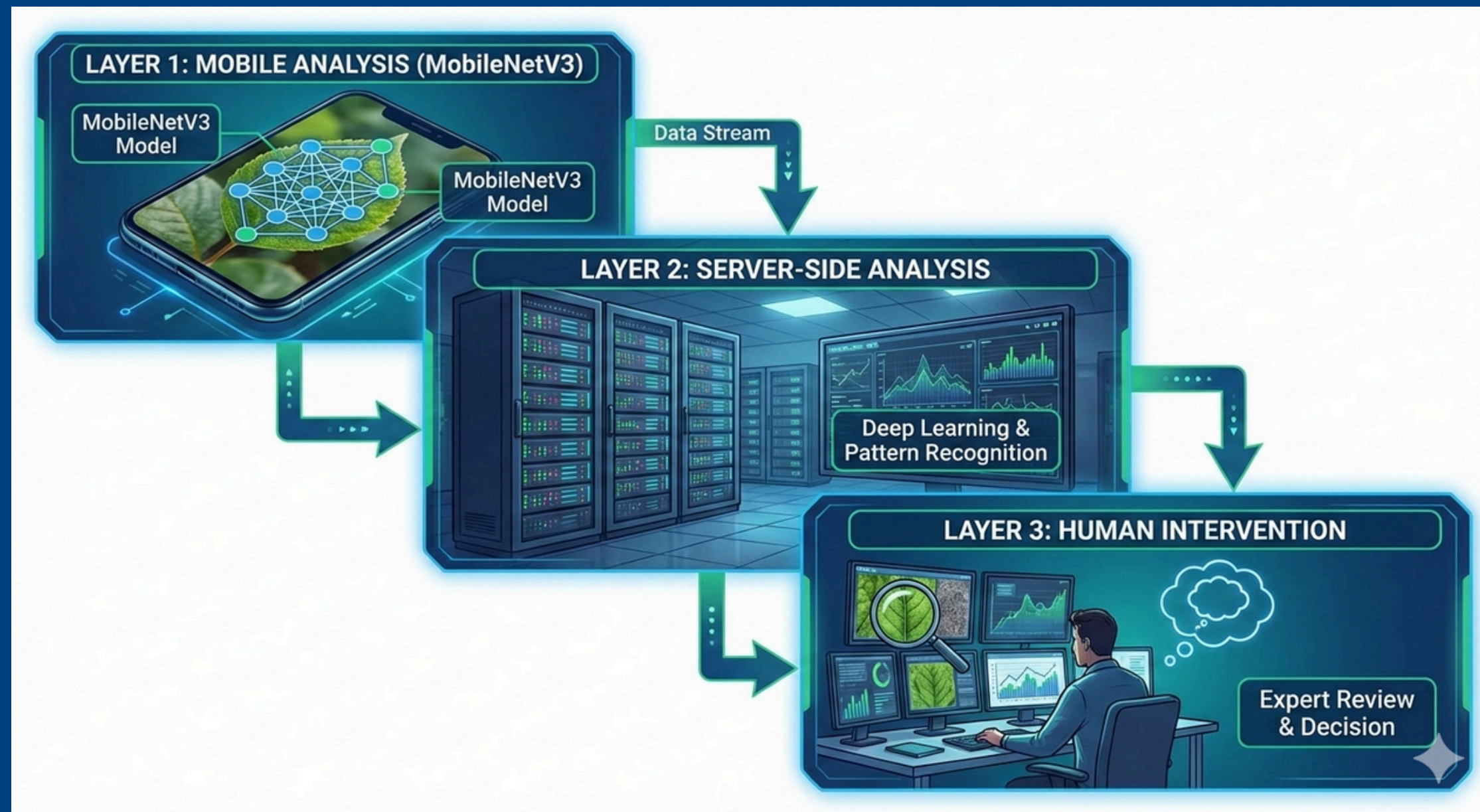
- **Passive Liveness**

- Analyzes facial texture, motion, and temporal consistency
- No explicit user action
- Better user experience and stronger security

Existing Top Solutions

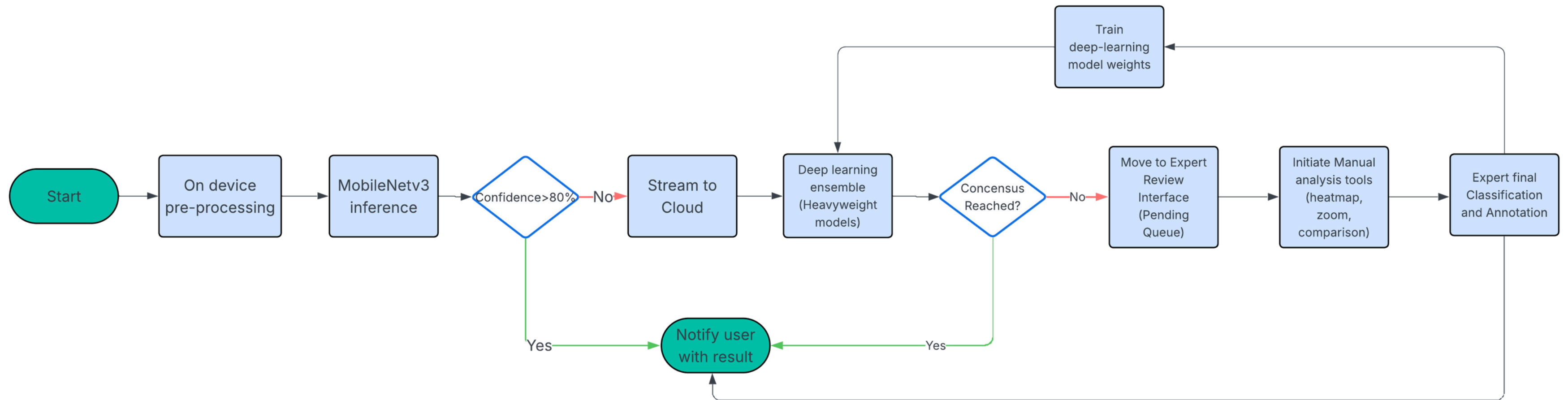
- Searching for MobileNetV3 (**ICCV 2019**)
- Applying Semantic Anchor in Face Anti-Spoofing Detection for Unified Physical-Digital Attacks (**ICCV 2025 Workshop**)
- Group-wise Scaling and Orthogonal Decomposition for Domain-Invariant Feature Extraction in Face Anti-Spoofing (**ICCV 2025**)
- Generalized Face Liveness Detection via De-fake Face Generator (**TPAMI 2024**)

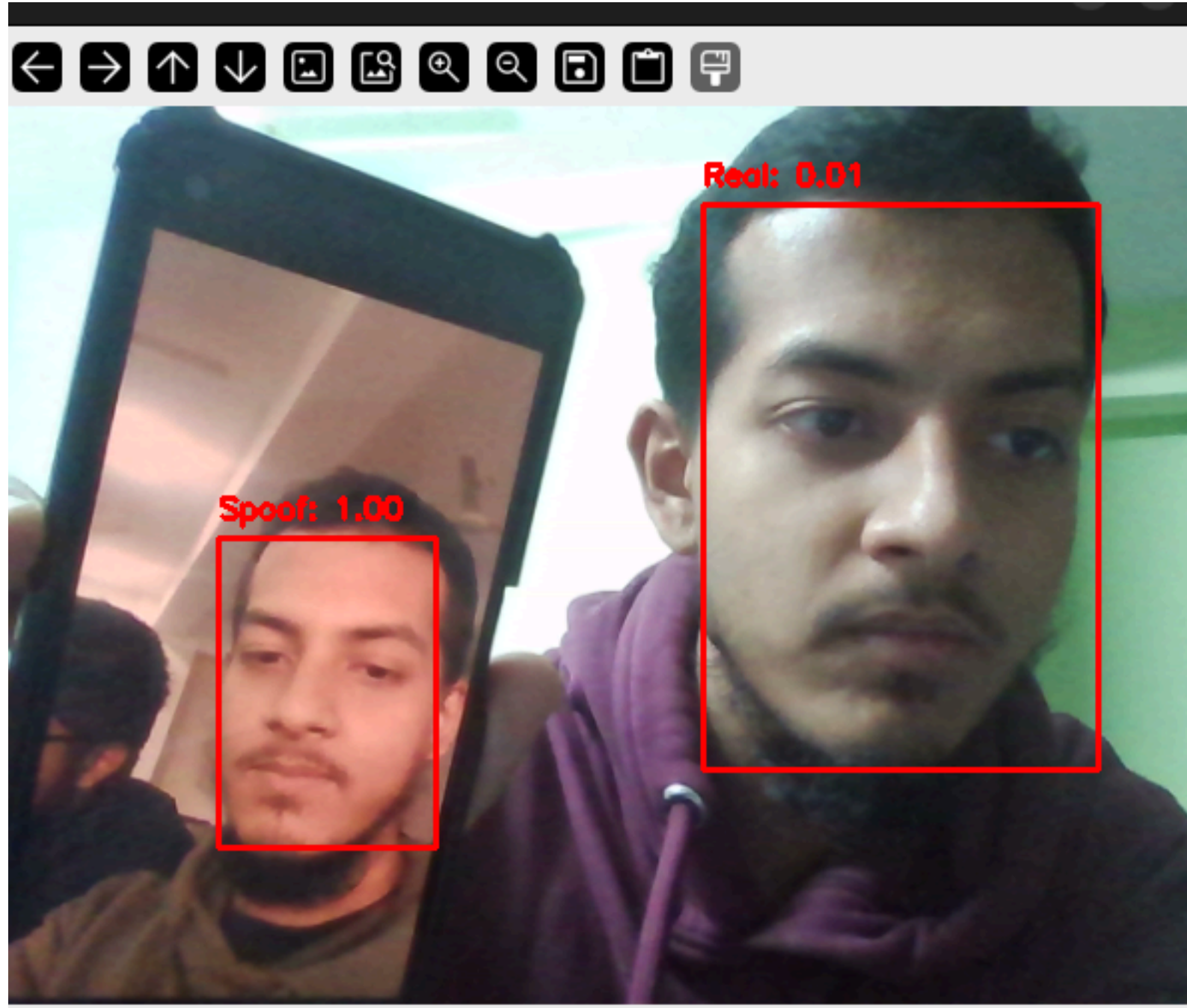
Solution Strategy



Multi-Layered
Defense
Framework

The complete flow





On-Device Detection

On-Device Detection

01

Frequency
Inconsistencies

Detects anomalies in image frequencies.

02

Micro-Texture
Irregularities

Identifies irregular patterns in surface texture.

03

Color Distribution
Anomalies

Analyzes color variations for authenticity.

Device Requirements

RAM for loading models :

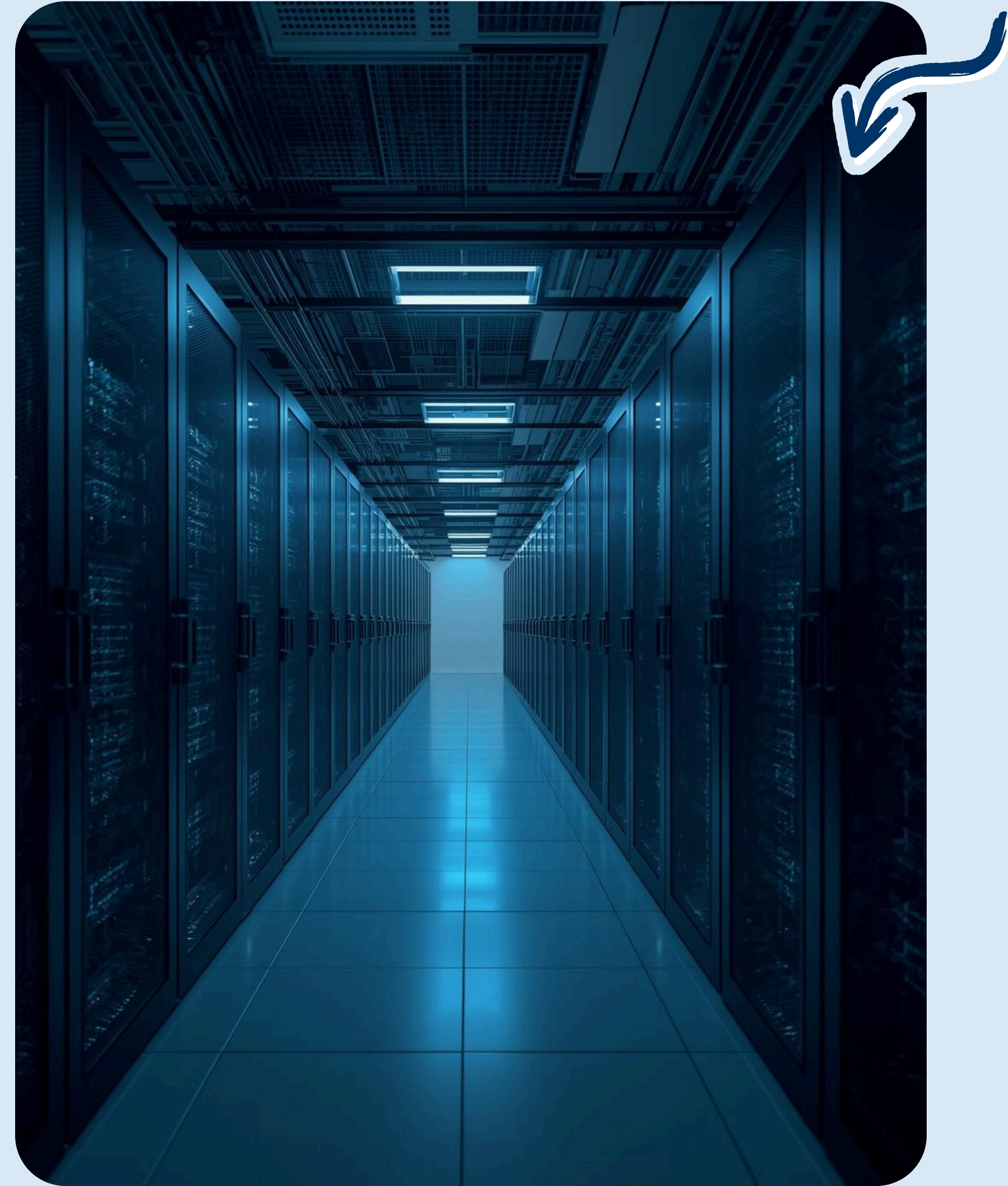
- Face detector (OpenCV SSD Caffe): ~15.9 MB
- Classifier (MobileNetV2 .hdf5): ~45.8 MB RSS
- Combined “models-only” additional RAM: ~62 MB

Inference : ~0.75 GB RAM for the inference process itself in steady state (CPU-only)

CPU: 4 physical cores / 8 threads (or better)

Latency : 25ms (~40fps)

Server-Side Deep Analysis



Server-Side Deep Analysis Overview



Global Contextual Analysis

This process examines features from a broader context, ensuring that the analysis considers multiple factors for accurate differentiation between real and synthetic faces.

Multi-Scale Image Evaluation

Advanced techniques evaluate images at various scales, allowing for the detection of inconsistencies that might not be apparent at a single resolution.

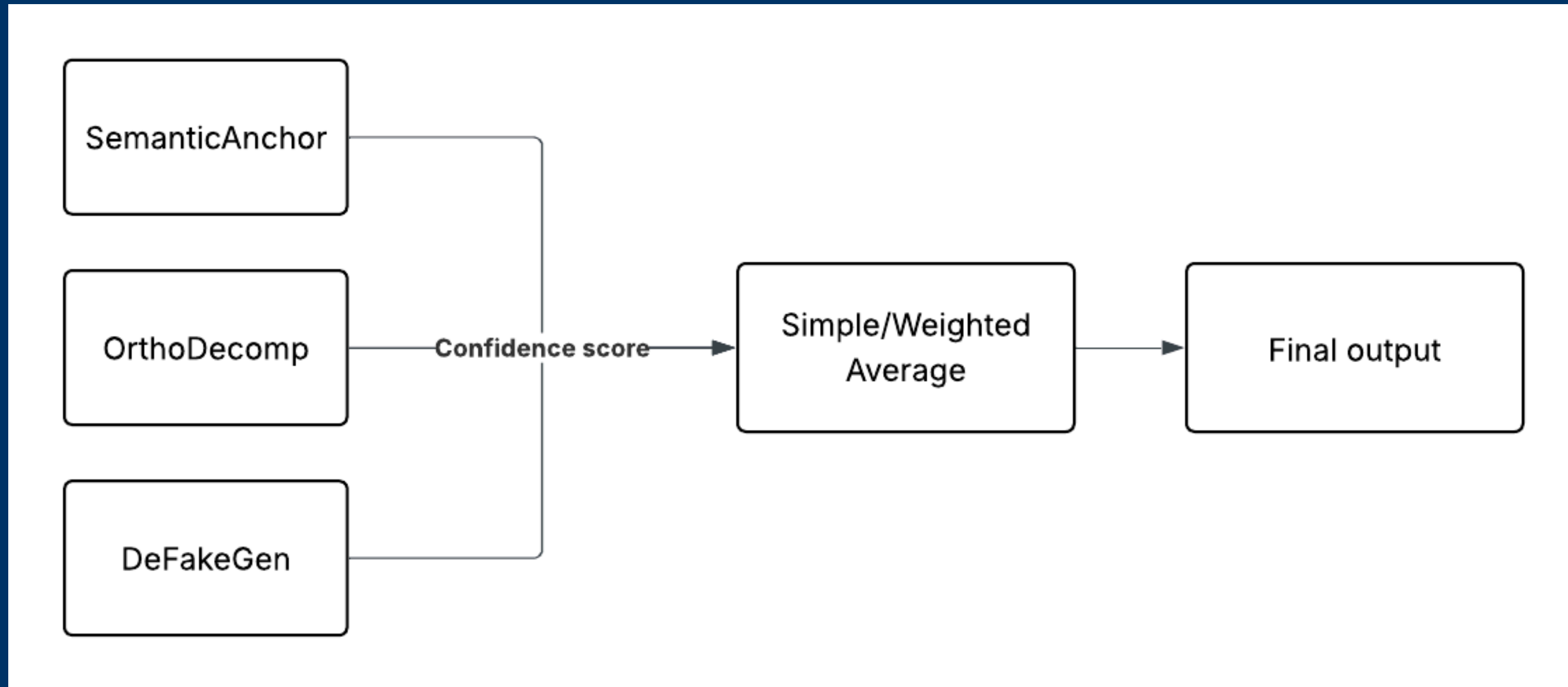
Spatial and Frequency Learning

Combines analysis of spatial information with frequency patterns to enhance the detection of subtle artifacts indicative of synthetic face generation.

Proposed methods

- **SemanticAnchor:** [Applying Semantic Anchor in Face Anti-Spoofing Detection for Unified Physical-Digital Attacks], ICCV 2025
 - Learns semantic alignment between visual features and descriptive text anchors for unified detection of physical and digital attack types
- **OrthoDecomp:** [Group-wise Scaling and Orthogonal Decomposition for Domain-Invariant Feature Extraction in Face Anti-Spoofing], ICCV 2025
 - Extracts domain-invariant features by orthogonal decomposition and group-wise scaling for better generalization to unseen spoof environments
- **DeFakeGen:** [Generalized Face Liveness Detection via De-fake Face Generator], TPAMI 2024
 - Transforms input into an estimated real face and uses residual anomalous cues as robust signals to detect fake or spoofed inputs

Ensemble



Datasets

- **CASIA-FASD**

- 600 video clips (150 live + 450 spoof)
- Warped printed photo, cut printed photo, video replay attacks

- **OULU - NPU**

- ~4,950 videos (real + spoof)
- Print, video replay (multiple environmental and device conditions)

- **Idiap Replay-Attack**

- ~1,300 video clips (real + spoof)
- Photo and video replay attempts under controlled/adverse lighting conditions

- **CelebA-Spoof**

- ~625,537 images
- diverse scenes, lighting, and devices

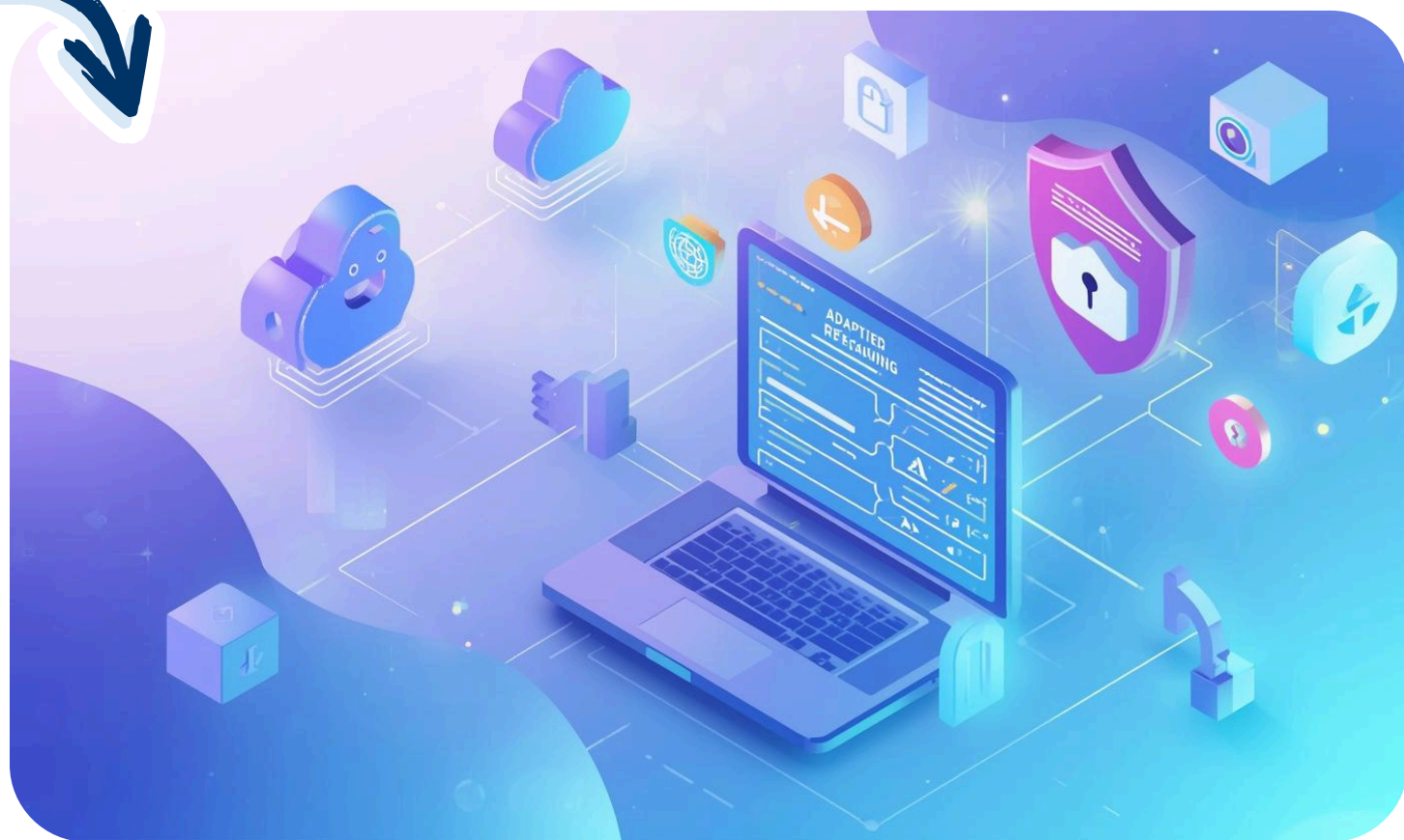
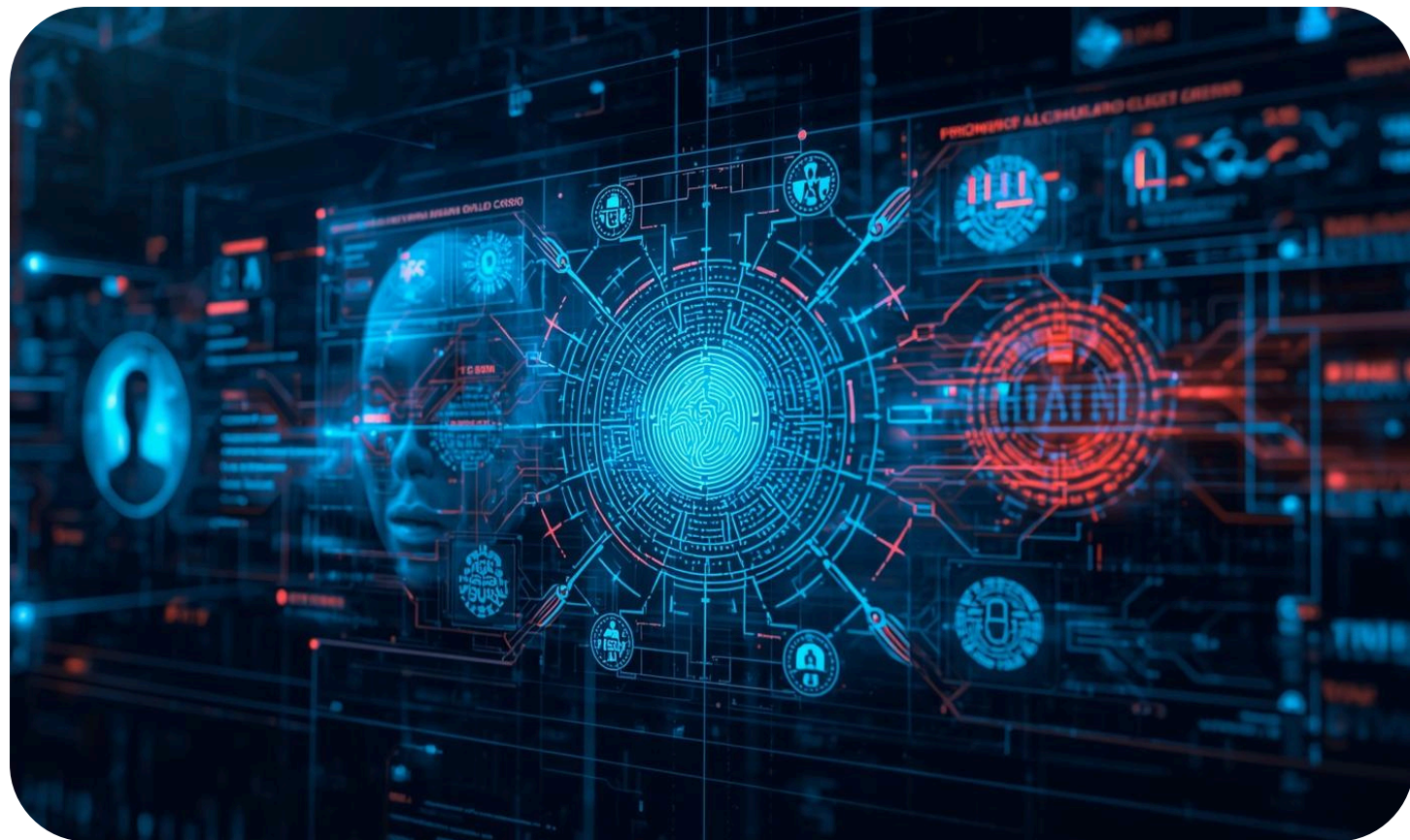
Evaluation Metrics

- **APCER** – Attack misclassification rate
- **BPCER** – Live user rejection rate
- **ACER** – Average PAD error rate
- **ROC-AUC** – Overall discrimination ability
- **EER** – Balanced threshold performance

Aligned with ISO/IEC 30107-3 PAD evaluation protocol

Extensions

- **Multimodal Analysis**
 - Prompt users to speak a phrase during verification
 - Analyze voice patterns along with facial cues for stronger identity verification
- **Additional Liveness Cues**
 - Prompt users to perform small facial actions: smile, blink, turn head, or follow on-screen gestures
 - Combines active and passive cues to detect sophisticated spoofing or replay attacks
- **NID/Image Cross-Verification**
 - Match the selfie captured in the liveness test with the user's official NID / ID document photo
 - Adds an extra layer of identity confirmation to prevent impersonation or fraud



Existing vs. Proposed Systems

Existing systems offer speed and widespread use but struggle with basic spoofing. In contrast, the proposed system leverages AI-specific detection, prioritizes automation, and enables adaptive learning for enhanced security.

Current System(MLKit+Human Review)

- **Strengths:**

- Detects simple spoofing (printed photos, replays, basic masks)
- Fast checks
- Widely used baseline

- **Weaknesses:**

- Cannot handle AI-generated images
- Misses advanced attacks
- Heavy human review: slow, costly, inconsistent
- Hard to scale
- No improvement as attacks evolve

Proposed Robust System

- **Strengths:**

- Built to spot AI-generated images
- Handles most cases without human review
- Multiple checks catch advanced attacks
- Can learn as new models appear
- Near-instant results
- Easy to update as technology changes



Key System Differentiators

Understanding the unique strengths of our approach

Our identity verification system stands out through its **proactive detection**, adaptive models, minimal human involvement, high scalability, and strong resilience against fraud, ensuring secure digital experiences.



Broader Implications

Economic Considerations

Key financial impacts of identity verification

Reduced Losses

Fraud prevention leads to significant financial savings.

Lower Costs

Automation minimizes operational expenses for verification processes.

Increased Confidence

Enhanced security boosts user trust in financial systems.

Sustainable Growth

Reliable verification fosters long-term financial stability and growth.



Environmental Considerations

Addressing energy consumption in AI systems

Energy Consumption

Server computations require significant energy resources.

On-Device Inference

Prioritizing local processing reduces energy use efficiently.

Performance Balance

Achieving efficiency without compromising system effectiveness.

Sustainable Practices

Implementing eco-friendly technologies promotes responsible operations.



Social Impact



Broader Access

Enhanced access to secure financial services for underserved communities and individuals.

Enhanced Trust

Building user confidence in digital identity verification through reliable systems and protocols.

Workforce Displacement

Automation may replace certain jobs, requiring workforce reskilling and adaptation strategies.

Risk of False Positives

Incorrect identity verification outcomes could exclude legitimate users from financial services.

Ethical Challenges

Balancing security requirements with user rights and ensuring fairness across demographics.

Ethical Considerations



Risks of Bias

Identifying biases in AI systems is crucial for fairness.

Diverse Data

Utilizing varied datasets helps mitigate demographic disparities in outcomes.

Transparency Reporting

Clear communication ensures users understand AI decision-making processes.

Consent-Based Usage

Obtaining explicit consent is essential for ethical data handling practices.

Accountability Measures

Establishing protocols for accountability strengthens trust in the system.

Ongoing Audits

Regular evaluations identify and correct biases in AI operations.

Thank You